Task-15:- Requirement Analysis

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for deploying a secure Nginx web server on AWS. The intended audience includes the **CTO, DevOps Team, and Security Architects** responsible for overseeing infrastructure integrity and data encryption.

1.2 Scope

- **Included:** Provisioning of AWS EC2, Nginx installation, OpenSSL certificate generation, UFW firewall configuration, and HTTPS verification.
- **Excluded:** Domain name registration (Route 53), Third-party CA integration (Let's Encrypt), and Automated CI/CD pipelines.

1.3 Definitions, Acronyms, and Abbreviations

- **SSL:** Secure Sockets Layer
- **TLS:** Transport Layer Security
- **UFW:** Uncomplicated Firewall
- **CSR:** Certificate Signing Request
- **EC2:** Elastic Compute Cloud

1.4 References

- AWS Well-Architected Framework - Security Pillar
- Nginx Official Documentation
- OpenSSL Man Pages

1.5 Overview

This document covers the project objectives, specific functional use cases (like certificate generation), security/performance constraints, and a risk analysis of using self-signed credentials.

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2. Project Overview

2.1 Project Objectives

- Deploy a functional web server with encryption in transit.
- Harden the OS against unauthorized network traffic.

- Validate the use of self-signed SSL for internal staging environments.

2.2 Business Goals

- Ensure zero-cost infrastructure using AWS Free Tier.
- Demonstrate technical competency in web security for ABC Technologies.

2.3 Background

The organization requires a baseline secure web server template. Current deployments are often HTTP-only; this project standardizes the move to HTTPS.

2.4 Constraints

- **Technology:** Must use Ubuntu Linux and Nginx.
- **Budget:** $0.00 (Must remain within Free Tier limits).
- **Certificate:** Must be self-signed for this specific phase.

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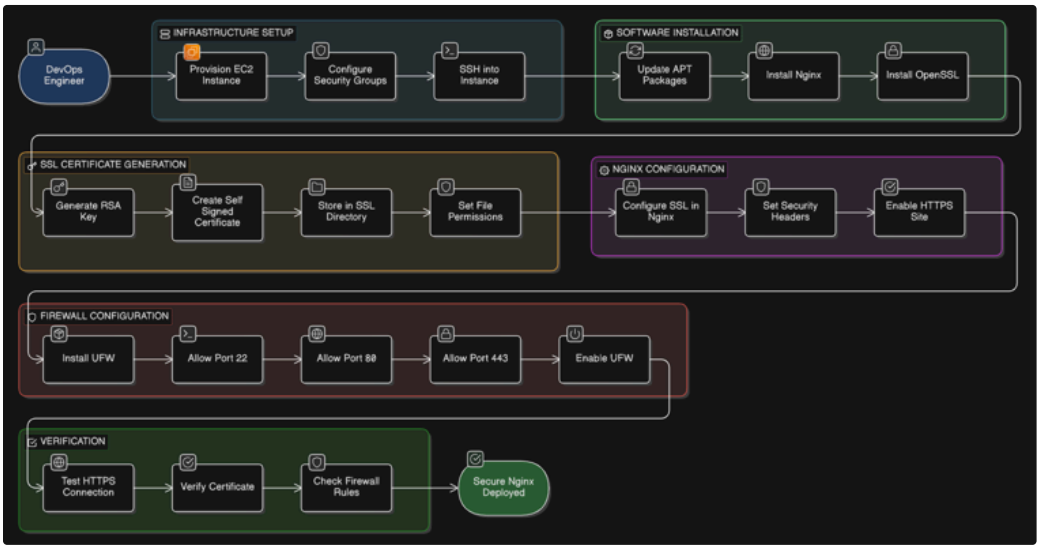
3. Functional Requirements

3.1 Use Case Diagram



3.2 Use Cases

3.2.1 Use Case 1: SSL Certificate Generation

- **Description:** Admin generates a 2048-bit RSA key and self-signed certificate.
- **Actors:** DevOps Engineer (Admin).
- **Preconditions:** OpenSSL is installed on the EC2 instance.
- **Postconditions:** `.crt` and `.key` files are stored in `/etc/nginx/ssl`.
- **Flow of Events:**
  a. Admin runs `openssl req` command.

b. Admin inputs organizational details (Surat, Gujarat, etc.).
c. System generates files and sets restricted permissions.

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4. Non-Functional Requirements

4.1 Performance Requirements

- **Response Time:** Web server should respond to requests in under 200ms.
- **Memory Footprint:** Nginx should utilize less than 5% of available system RAM.

4.2 Security Requirements

- **Encryption:** Use TLS 1.2 or higher.
- **Firewall:** Only ports 80, 443, and 22 should be open.
- **Headers:** Implement `X-Frame-Options: SAMEORIGIN`.

4.4 Reliability Requirements

- **Uptime:** 99.9% availability of the Nginx service.
- **Recovery:** Systemd must automatically restart Nginx on process crash.

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5. Assumptions and Dependencies

5.1 Assumptions

- The user has valid AWS credentials and an active SSH key pair.
- The `t3.micro` instance has outbound internet access for package updates.

5.2 Dependencies

- **APT Repositories:** Dependent on Ubuntu official mirrors for Nginx binaries.

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6. Risk Analysis

6.1 Risk Identification

- **Risk:** Browsers display a "Connection not private" warning.
- **Impact:** Medium (decreased user trust).

6.2 Risk Mitigation

- **Strategy:** Include a "Help" page or documentation explaining that the internal environment uses self-signed certificates for encryption testing.

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7. Glossary

- **Self-Signed Certificate:** An identity certificate that is signed by the same entity whose identity it certifies.
- **Hardening:** The process of securing a system by reducing its surface of vulnerability.

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9. Customer Acknowledgement

Name: Pushkar Khatavkar

Title: DevOps Intern / Engineer

Signature: _____

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